

SMAAs, EMAAs, or Other?

“Everyone’s children are above average,” said Tom meanly.

Moving averages are discussed in this chapter simply because they are ubiquitous in technical analysis. There are many better filters for various applications that are described in the following chapters. Moving averages have the advantage that they are dirt-simple to compute. Their disadvantage is that they produce only a small amount of smoothing for the cost of lag in their computation.

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One little-known fact is that a simple moving average constitutes a best fit to data in the least-squares sense.

■ Simple Moving Averages (SMAAs)

SMAAs are a special case of nonrecursive filters where all the coefficients are equal to unity divided by the number of elements in the filter. At any given point in the data stream, the output of the filter is just the average of data over the span of the filter. When time is advanced by one bar in one algorithm, the fractional part of the oldest data sample is discarded and the fractional newest data sample is added. Thus, it is a “moving” average. It just doesn’t get more simple than that, and so it is deemed to be a simple moving average.

Viewed from another perspective, it can be viewed as a “window” that is passed across a fixed data stream. Only those data samples within the window are averaged and provided as an output of the filter. Since modern computers are very fast, I recommend this latter approach because if the

length of the average is varied across the data set, the computations will fail. We may want to make the length of the average vary when adapting the filter to volatility or a measured cycle period.

An impulse is a theoretical data object that has zero width and infinite height. It is used to assess the filter transfer response. For example, you can think of it as a clapper striking a bell—the bell provides the transfer response, and the ringing is the output. In other words, the ringing is the impulse response of a bell. In sampled data systems, we can approximate an impulse as a data stream that has a value only at one point and has zeros at all other data samples.

If we slide the SMA window across the impulse data stream, the filter will have an output only when the impulse falls within the window of the filter. Thus, the impulse response of an SMA is finite, and therefore an SMA is a special case of a finite impulse response (FIR) filter.

Suppose we start a filter design by trying to make its transfer response be the best fit to a straight line. If we have more than two data points, this would be impossible in the general case. However, we can define a best fit, where the sum of the squares of the differences between the data and our straight line is minimum. In the case of five data points, designated as u_x , we can write the desired function as

$$F = \sum_{x=-2}^2 [u_x - (mx + b)]^2$$

We can minimize this function by taking the derivatives of this function with respect to m and b and setting those derivatives to zero. When we do this, we find that the correct value of b is just the average of the five data points. This is true regardless of which five data points we select, so that an SMA is synonymous with finding a best-fitting straight line to the data, at least when the lag of the filter is disregarded. I find this to be a fascinating little-known fact about SMAs.

In Chapter 1, we showed that the lag of a nonrecursive filter having coefficients symmetrical about its center point and of length N is exactly $(N - 1) / 2$. Since an SMA is a special case of a nonrecursive filter, it is also true that an SMA lag is also

$$(N - 1) / 2$$

Equation 1-8 is repeated here as the amplitude of the transfer response of an SMA.

$$H(2\pi f) = \frac{\sin(N\pi f)}{N\sin(\pi f)} \quad (2-1)$$

At zero frequency, Equation 2-1 evaluates to 1 in the limit. This starts the pass band of the SMA. The critical period of a filter output is the frequency at which the output power of the filter is half the power of the input wave at that frequency. Since power is proportional to the square of the wave amplitude, we are interested in finding the frequency at which the output wave has an amplitude of 0.707 (the square root of 0.5). Knowing this, we can select an SMA that gives us the approximate critical period we seek. Figure 2.1 shows the relationship between the critical period and the number of elements in the SMA.

It is a reasonable approximation and easy to remember that the critical period of an SMA is twice its length.

The first zero in the transfer response occurs when the argument in the numerator has a value of π . This occurs when $f = 1 / N$. Subsequent zeros in the transfer response occur when the argument in the numerator are integer multiples of π . The first maximum in the stop band occurs when the argument in the numerator causes the sine function to be at a maximum, or $f = 3\pi / 2$. At this frequency the value of the transfer response is approximately $2 / (3\pi)$, or -13.3 decibels (dB). For the same reason, the next maximum in the stop band occurs at $f = 5\pi / 2$, and the attenuation is $2 / (5\pi)$, or -17.8 dB. This amount of attenuation is not substantial, and you will see methods to modify an SMA to improve its filtering capability.

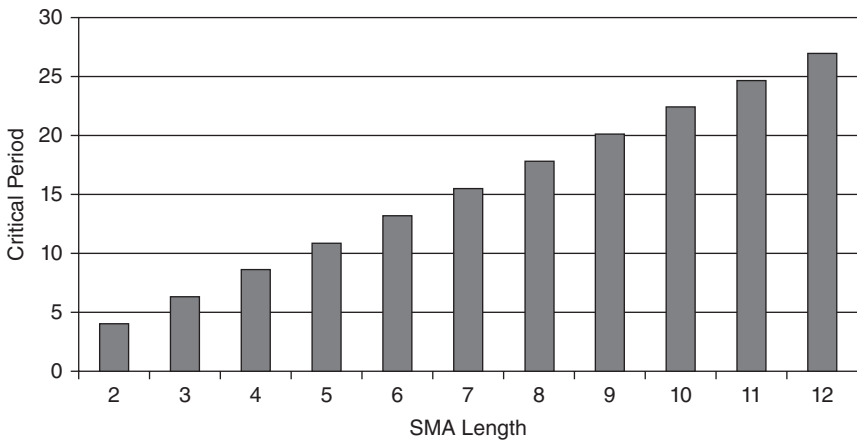


FIGURE 2.1 SMA Critical Period as a Function of Filter Length

■ Exponential Moving Averages (EMAs)

An EMA is computed by taking a fraction of the current price added to the complementary fraction of the EMA calculation one bar ago. The first fraction is usually called alpha, and can have a value between zero and one. The two fractions must sum to unity, so the second fraction must have the value of one minus alpha. The equation to compute an EMA is:

$$\text{EMA} = \alpha * \text{Price} + (1 - \alpha) * \text{EMA}[1] \quad (2-2)$$

The EMA becomes a moving average by moving from bar to bar, from left to right, across the price data.

Imagine a case where the input data set has only one sample at an amplitude of $1 / \alpha$. When the EMA is applied to these data, the first output from the filter is unity because there was no previous value for the EMA. On subsequent calculations, the price value is zero, and so the sequence of calculations is:

$$\text{EMA}(0) = 1$$

$$\text{EMA}(1) = (1 - \alpha)$$

$$\text{EMA}(2) = (1 - \alpha) * (1 - \alpha) = (1 - \alpha)^2$$

$$\text{EMA}(3) = (1 - \alpha)^2 * (1 - \alpha) = (1 - \alpha)^3$$

...

$$\text{EMA}(N) = (1 - \alpha)^N$$

Since the quantity $(1 - \alpha)$ must be less than one, the amplitude decays as the exponent of each succeeding calculation from an impulse input, hence the name exponential moving average. In principle, a part of any data input remains in subsequent calculations, although the contribution becomes vanishingly small. This attribute makes an EMA part of a general class of filters called infinite impulse response (IIR) filters.

It is instructive to examine the EMA response to a step function. A step function has a series of constant values and then jumps to another series of constant values. Assume the price has been zero for a long time and then suddenly jumps up to a value of 1 and maintains that value thereafter. On the first bar, the EMA will have a value of α . On the second bar, the value will be $\alpha + \alpha * (1 - \alpha)$. On the third bar, the value will be $\alpha + \alpha * (1 - \alpha) +$

$\alpha * (1 - \alpha)^2$, and so on. The EMA will gradually approach the value of 1. A common error in programming is to insert a value for α , such as .2, and inserting another number for $(1 - \alpha)$, such as .9. The two terms must sum to unity or the recursive algorithm will lead to erratic results or might even cause your computer to crash. You should always check your computer code to ensure that the two terms sum to unity. I am so cautious on this point that I assign the value α as a global variable and write out the EMA equation in terms of α . By letting the computer do the work, I know the two terms must sum correctly.

We can assert that the price on day “d” is d . If we assume the lag of the EMA is “L,” then the current value of the EMA is $(d - L)$. Further, the previous EMA would have a value of $(d - L - 1)$, since price is rising one unit per bar. Putting these values into the equation for the EMA, we obtain

$$\begin{aligned}(d - L) &= \alpha * d + (1 - \alpha) * (d - L - 1) \\ &= \alpha * d + (d - L) - 1 - \alpha * d + \alpha * (L + 1) \\ 0 &= \alpha * (L + 1) - 1 \\ \alpha &= 1 / (L + 1)\end{aligned}$$

This equation shows that we can select an acceptable lag and, from that lag, compute the alpha term of the EMA. For example, if we can accept a three-bar lag resulting from the EMA, we would use $\alpha = 0.25$.

We can also relate an EMA to an SMA on the basis of their equivalent static lags. Recalling that the lag of an SMA is $(N - 1) / 2$ for an N -bar SMA, we can substitute this value of lag into the alpha calculation of the EMA as

$$\begin{aligned}\alpha &= 1 / ((N - 1) / 2 + 1) \\ &= 2 / ((N - 1) + 2) \\ &= 2 / (N + 1)\end{aligned}\tag{2-3}$$

This is the relationship between an N -bar SMA and the alpha of an EMA that is quoted in most technical analysis books. The frequency responses of a five-bar SMA and its EMA equivalent are compared in Figure 2.2. Figure 2.3 shows that the lags of the SMA and EMA are equal only at zero frequency.

Although the lags at static frequencies align, note that the critical period of the EMA is not aligned with that of the SMA. We can obtain a better calculation for the fraction α by examining the transfer response of the EMA.

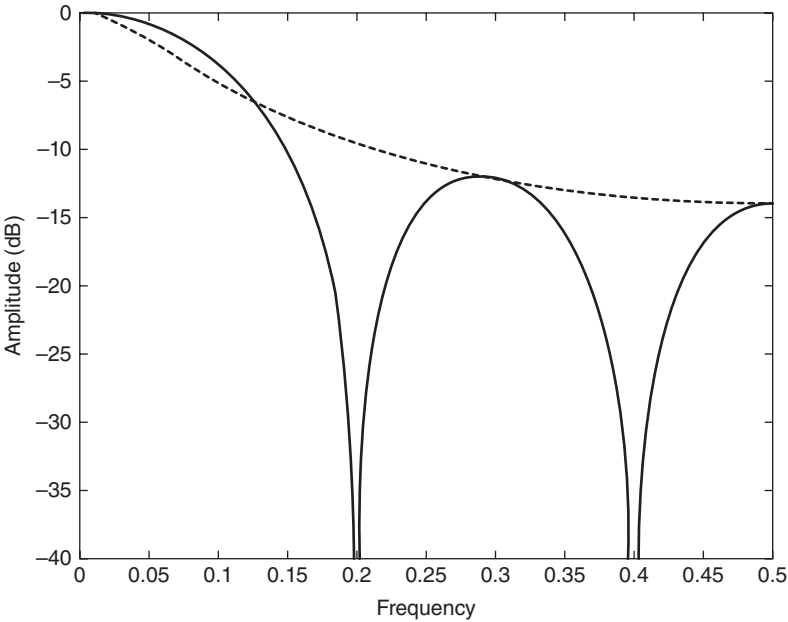


FIGURE 2.2 Frequency Response Comparison of a Five-Bar SMA and Similar EMA

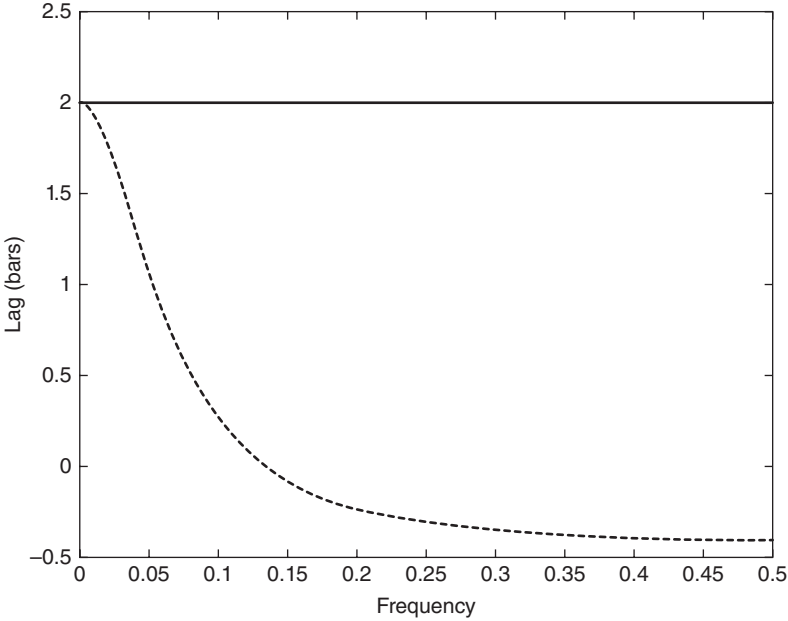


FIGURE 2.3 Lag Response of EMA Compared to a Constant Two-Bar Lag of a Five-Bar SMA

■ Weighted Moving Averages (WMAs)

It appears that the WMA was invented by a trader who did not have a firm grasp of filter theory in hopes of reducing lag.

A WMA is an FIR filter having coefficients asymmetrically weighted about the center point of the filter. The most recent data sample is weighted most heavily, the second most recent data sample weighted somewhat less, and so on. The weighting does not necessarily have to be linear with the data sample, but that is the most common usage. For our graphical example, I will use a five-element WMA whose coefficients are $[5\ 4\ 3\ 2\ 1] / 15$. Figure 2.4 shows the frequency response of this filter, and Figure 2.5 shows its lag response.

Figure 2.4 shows that a WMA has poorer attenuation than the equivalent SMA shown in Figure 2.2 and that a WMA has more lag in the pass band than the equivalent EMA shown in Figure 2.3. In short, a WMA has little to recommend itself compared to the other common moving averages.

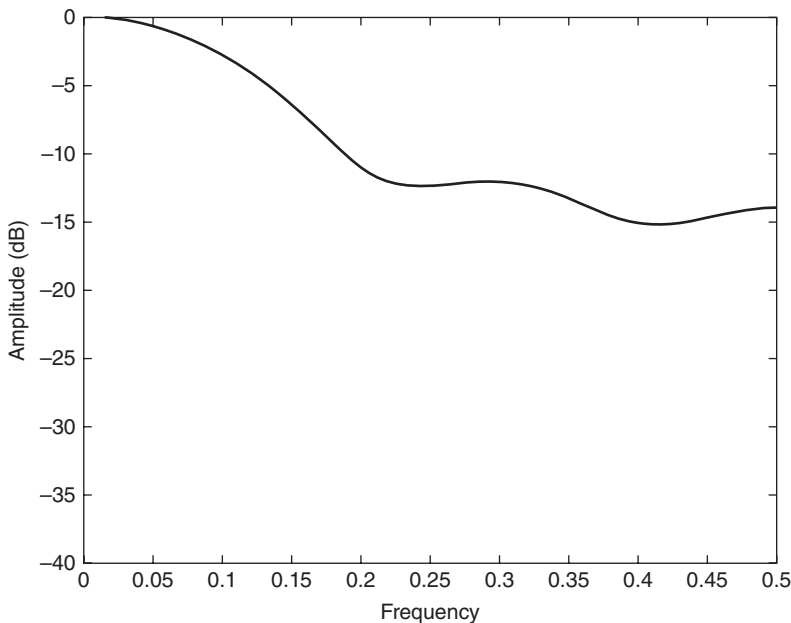


FIGURE 2.4 A WMA Has Poor Rejection in Its Transfer Response

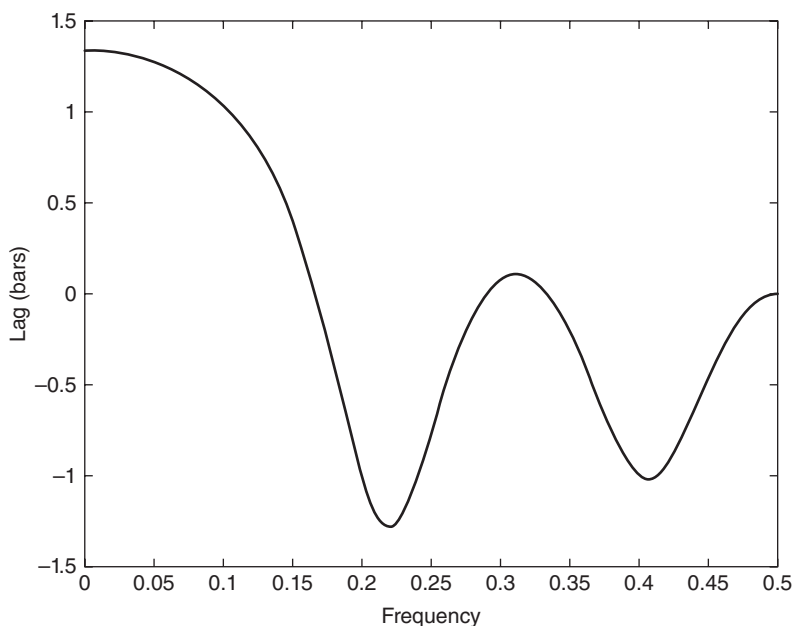


FIGURE 2.5 Lag of a Five-Element WMA Has No Outstanding Characteristics

■ Median Filter

There are times when a median filter is the most appropriate filter to use. It is best applied when the data contains impulsive noise or when there are wide variations in the data, such as volume data. Its purpose is to attempt to discard outlier data samples to avoid having them exert an inordinate impact on the average.

A median filter simply picks the middle value of a sample set of data after the samples have been ranked. This selection is easiest when an odd number of data samples is used. To demonstrate the effect of a median filter, consider a data set consisting of four samples, each having a value of 1 and one sample that has a value of 6. The data samples can appear in any order. After ranking, the center value is just 1. Therefore the median value is 1, which happens to be the value of most of the data samples. However, the average value is $(4 + 6) / 5 = 2$. That one outlier data sample has severely skewed the average, and the median filter best recovers the “average” value of the data set.

■ Key Points to Remember

1. The critical period defining the pass band of an SMA is approximately twice the length of the filter.
2. An N -length SMA will have a lag of $(N - 1) / 2$ bars at all frequencies.
3. An EMA should always be written as $EMA = \alpha * Input + (1 - \alpha) * EMA[1]$ to avoid computational errors in evaluating the fractional coefficient.
4. The fractional coefficient of an EMA equivalent to an N -length SMA is calculated as $\alpha = 2 / (N - 1)$.
5. A WMA has little or no redeeming virtue.
6. A median filter is best used when the data contain impulsive noise or when there are wild variations in the data. Smoothing volume data is one example of a good application for a median filter.